Highlights

skycell: a cost-based HEALPix index for sky positions in PostgreSQL

Jesús Salgado, Michele Delli Veneri

- A HEALPix B-tree index for PostgreSQL answering cones, polygons and cross-matches
- A cost model sets each covering's depth from local density and index range price
- Faster than pgSphere at every cone radius from 1 arcsec to 3 deg, real Gaia DR3 too
- Cross-matches 20–33% faster than q3c_join, from an index a third of pgSphere's size
- Speaks ADQL; one column holds circles and polygons, read and written as IVOA STC-S

skycell: a cost-based HEALPix index for sky positions in PostgreSQL

Jesús Salgado[a,*], Michele Delli Veneri[a]

[a]SKA Observatory, Jodrell Bank, Lower Withington, Macclesfield, SK11 9FT, United Kingdom

Abstract

Astronomical archives answer positional queries from relational databases, and in PostgreSQL two extensions dominate: Q3C, which keys sources by a cube quad-tree index in a B-tree, and pgSphere, which stores spherical geometry in a GiST tree and also indexes regions. We present `skycell`, a PostgreSQL extension that keeps the integer key and the B-tree yet answers cones, polygons, cross-matches and stored footprints in the archives' own ADQL vocabulary. Positions are keyed by their order-29 HEALPix cell, and a query region is covered by cells whose depth is chosen by a cost model: the price of an extra index range against the false-positive rows it removes, with local source density read from the histogram PostgreSQL already keeps. Cones and cross-matches are answered by a planner custom scan that walks the covering's ranges, polygons by a rewrite into range conditions, and stored regions by a GiST operator class. Coverings are validated by brute force over millions of points. In randomized paired trials on PostgreSQL 18.6 with 10- and 50-million-row catalogues, including real Gaia DR3 positions, `skycell` is faster than pgSphere at every cone radius from 1″ to 3°, by 4–33% with a warm cache and up to 2.7 times cold. It cross-matches 20–33% faster than Q3C's `q3c_join` and three times faster than pgSphere, and answers convex polygons 2.7–4.4 times faster than pgSphere, from an index a third the size. One index serving points, regions and cross-matches through ADQL makes `skycell` a strong default for an archive's positional layer.

Keywords: Astronomical databases, Spatial indexing, HEALPix, PostgreSQL, Cross-matching, Virtual Observatory

1. Introduction

Most questions asked of an astronomical archive are positional: which sources lie in this cone, which observations cover this field, which catalogue sources match my target list. In the Virtual Observatory they arrive as ADQL (Ortiz et al., 2008) through the Table Access Protocol (Dowler et al., 2019), usually generated by a service layer (e.g. Demleitner et al., 2014), and at the scale of the ESA Gaia Archive (Salgado et al., 2017, 2020), which serves about 10^9 sources and offers server-side cross-matching.

For those who run such archives the questions are practical: how fast cone searches and cross-matches are on a large catalogue, how large the index is, how long it takes to build and how much memory it needs, whether performance holds in crowded fields and at scale, and how much an existing database has to change. In PostgreSQL the usual answers are two extensions. Q3C (Koposov & Bartunov, 2006) stores each position as a 64-bit integer from a quad-tree on a cube in a B-tree: compact and fast for points, with `q3c_join` for cross-matching. pgSphere (Chilingarian et al., 2004) stores spherical geometry in a GiST tree: more general, and able to index stored regions such as observation footprints. Archives that need both kinds of query often run both. Related designs include the Hierarchical Triangular Mesh (Kunszt et al., 2001) and the zones algorithm (Gray et al., 2006).

This paper presents `skycell`, a PostgreSQL extension that covers both roles with one index and ADQL's own vocabulary. Positions are stored as an integer HEALPix cell in an ordinary B-tree, as in Q3C, but how a query region becomes index ranges is chosen per query by a cost model from statistics PostgreSQL already keeps. Stored regions get their own GiST index. Section 2 explains how both indexes work, Section 3 why they do not miss rows, Section 4 how to use them in an existing archive, Section 5 how they compare with Q3C and pgSphere, and Section 6 when `skycell` is worth adopting.

2. How skycell works

2.1. Positions as integers

A B-tree can only answer range conditions on a sortable key, so a position on the sky has to become a number first. `skycell` uses the position's HEALPix cell at order 29 in the nested scheme (Górski et al., 2005): a 64-bit integer, with cells about 0.4 mas across. Three properties of this key matter. First, cells are *nested*: all the order-29 cells inside a larger cell p at order k form one contiguous run of integers,

$$[\, p \cdot 4^{29-k},\ (p+1) \cdot 4^{29-k} - 1 \,], \tag{1}$$

so any region covered by cells of any size becomes a set of integer ranges that the B-tree answers with range scans. Second, cells have *equal area*, so the number of keys in a range is proportional to the source density in that part of the sky; the histogram PostgreSQL's `ANALYZE` already keeps for the key is

*Corresponding author

Email addresses: `jesus.salgado@skao.int` (Jesús Salgado), `michele.delliveneri@skao.int` (Michele Delli Veneri)

therefore a map of crowding (Section 2.2). Third, the numbering is *standard*: Gaia's `source_id` (Gaia Collaboration, 2016) and IVOA MOC maps (Fernique et al., 2014) use it, so keys and footprints can be exchanged with other tools.

2.2. Choosing the resolution of a covering

To answer a cone search, `skycell` covers the cone with cells, reads the index ranges those cells correspond to, and removes rows that fall inside the covering but outside the cone with an exact test. The cell size is a trade-off: large cells give few index ranges but many false candidates to fetch and reject, while small cells follow the edge closely but each extra range costs an index descent. The balance depends on crowding: in an empty field false candidates are cheap, while in the Galactic bulge it pays to cut finer.

`skycell` makes this choice with a cost model. For cells of size s covering a region of radius r where the local source density is ρ,

$$C(s) = n_{\mathrm{ranges}}(s)\, c_{\mathrm{range}} + \rho\,(A_{\mathrm{cover}}(s) - A_{\mathrm{region}}), \tag{2}$$

where both terms are measured in rows: the second is the expected number of false candidates, and c_{range} is the cost of one extra range expressed as the number of false candidates it is worth to avoid it. When the cells are much smaller than the region, the covering follows the boundary, so the number of ranges grows as r/s and the wasted area as the perimeter times s. Writing $A_{\mathrm{cover}} \simeq \pi(r+\beta s)^2$ and $n_{\mathrm{ranges}} \simeq \gamma r/s$ and minimising Eq. (2) gives

$$s_\star = \sqrt{\frac{\gamma\, c_{\mathrm{range}}}{2\pi\beta\,\rho}}, \tag{3}$$

which does not depend on the size of the region: only the density and the range price decide how finely to cut. Since the approximation fails for cells as large as the region, the implementation evaluates Eq. (2) at all 30 orders and takes the cheapest. For example, a 0.25° cone holding 11 sources at high Galactic latitude gets one range of large cells (5.6 times the cone's area); in the bulge, holding 1183 sources, it is cut two orders finer (8 ranges, 1.3 times the area). Two guards protect against bad inputs: a partly covered cell may not exceed 64 times the region's area, and neighbouring ranges are merged when the rows between them cost less than the extra range.

The range price.. c_{range} needs no tuning: it is the ratio of two costs PostgreSQL already models. A range costs a B-tree descent plus two pages (the index leaf it lands on and the heap page where reading stops being sequential); a false candidate costs its index entry, its heap tuple, the exact test and its share of a heap page. Pricing each page by where it is likely to come from, $p = f\,50\,c_{\mathrm{op}} + (1-f)\,c_{\mathrm{page}}$, gives

$$c_{\mathrm{range}} = \frac{\lceil \log_2 N \rceil\, c_{\mathrm{op}} + 50(H+1)\,c_{\mathrm{op}} + 2p}{c_{\mathrm{tuple}} + c_{\mathrm{itup}} + k\,c_{\mathrm{op}} + p/n_{\mathrm{pp}}}, \tag{4}$$

where N is the number of rows, H the B-tree height, n_{pp} the rows per heap page, k the operator count of the exact test, the c terms PostgreSQL's own planner cost settings, and $f = \min(1, \mathrm{shared_buffers}/\mathrm{relpages})$ the fraction of the table the buffer pool can hold. For a 10-million-row catalogue this gives 34 rows when the table fits in memory and 87 when it does not; a table with wide rows, and so fewer rows per page, is cut finer. The exact value matters little: the median query time changes by less than 25% for c_{range} between 3 and 300, and the order chosen is within about 3% of the best.

The density estimate.. The density ρ comes from the histogram `ANALYZE` keeps for the indexed key: $B+1$ bounds $h_0 < \cdots < h_B$ such that each bucket holds about N/B rows. Because the key has equal-area cells, the rows in a key range $[a, b]$ can be estimated by linear interpolation within the buckets it spans,

$$\hat{n}(a,b) = \frac{N}{B} \sum_i \frac{\big|[a,b] \cap [h_i, h_{i+1})\big|}{h_{i+1} - h_i}, \tag{5}$$

which gives a density in rows per steradian of

$$\hat{\rho}(a,b) = \frac{\hat{n}(a,b)}{(b-a+1)\,A_{29}}, \tag{6}$$

with A_{29} the area of a leaf cell. The estimate is taken over the cell containing the region's centre, at the order whose cells are about four times the region's size. A cluster smaller than one histogram bucket is averaged away, which the area guard keeps in check; Section 5.6 measures the estimate on real Gaia positions.

Never dropping a cell that matters.. While building a covering, each cell is classified as inside the region, outside it, or on its boundary. Only one of these decisions can lose data: a cell wrongly called *outside* is dropped, and its rows silently disappear from the answer, while a wrong *inside* or *boundary* decision only adds candidates that the exact test removes. The outside decision therefore uses a strict bound on how far a cell's points can lie from its centre,

$$R_{\mathrm{cell}}(k,p) = \min\big(\mathrm{maxpixrad}(k),\ \max_i \angle(\mathbf{c}_{k,p}, \mathbf{v}_i)\big), \tag{7}$$

where $\mathbf{c}_{k,p}$ is the cell centre, $\mathbf{v}_i$ its corners, and $\mathrm{maxpixrad}(k)$ the analytic worst case of the reference HEALPix implementation. A cell is outside a cone of centre $\mathbf{q}$ and radius r only if $\angle(\mathbf{q}, \mathbf{c}_{k,p}) - R_{\mathrm{cell}} > r + \epsilon$ (for a polygon, the same test against each bounding half-space), with a margin $\epsilon = 2 \times 10^{-13}$ rad for floating-point error, five orders of magnitude below the leaf cell size. The inside test can be less strict, since an error there costs time, not rows.

2.3. Answering queries in PostgreSQL

The extension is written in C and plugs into the PostgreSQL planner through support functions on its spatial predicates. A cone – written as `skycell_cone`, as the Q3C-style `skycell_join` or `skycell_radial_query`, or as ADQL's `point <@ circle` – is handled by a planner *custom scan*, a scan node that `skycell` adds to PostgreSQL. The planner considers it wherever a B-tree on the cell key can serve the query, and costs it as it would cost the same work done by built-in

scans. At execution the node computes the covering and reads its ranges in one of two ways, chosen from the table's physical order as PostgreSQL chooses between an index scan and a bitmap scan: one index scan per range when the table is stored in cell order, or all ranges into one bitmap, reading each heap page once, when it is not. Every row is then checked with the exact test.

A cross-match is a cone whose centre comes from another table's row. The same node then runs on the inner side of a nested loop and computes a fresh covering for each target, with as many ranges as that target's cone needs. Its row estimate is measured at plan time on 32 sampled targets, because a density model cannot know that cross-match targets usually sit on catalogue sources. Polygons and boxes take a simpler path: before planning, the predicate is rewritten into range conditions on the key plus the exact test, which the planner then estimates from the same histogram. Coverings, densities and index lookups are cached per database session.

2.4. Stored regions

Archives also store regions, such as the observation footprints of an ObsCore `s_region` column, and ask which footprints contain a position or overlap a region. `skycell` stores them in one type, `skyregion`, whose values can be circles or polygons, so one column holds both; pgSphere needs a separate type, and so a separate column, for each.

A `skyregion` column takes a GiST index, which serves overlap (`&&`), containment of a position, and containment between regions. GiST stores a simplified key for each region and organises the keys in a tree, each internal entry bounding all the keys below it. A query descends only into entries whose key could match and then tests the real geometry of each candidate. The default key, operator class `skyregion_box4_gist_ops`, is an axis-aligned box in 3D Cartesian coordinates around the region, stored in single precision (24 bytes, the representation pgSphere uses). It is small and cheap to compare, but loose around a large circle. The second operator class, `skyregion_gist_ops`, keys each region by a few spherical caps, exact for a circle at any radius; its larger keys make it slower on regions of similar size, but better for a column mixing very different scales, such as survey tiles next to small footprints. Footprints can also be stored as MOC cells (Fernique et al., 2014) in an ordinary B-tree (or a GIN index over each region's cells, which the planner uses automatically when present): the footprints containing a position are then those whose cell is one of the position's ancestor cells, a short list of exact lookups.

3. Correctness

An index that misses a row gives a wrong answer with no warning. Both index paths therefore approximate only in the direction that adds candidates, and the exact geometry decides; this section states what that rests on.

Cone and polygon coverings (B-tree).. The covering is complete:

> **Proposition.** Let R be a cone or convex polygon and let C be the covering computed for it. If (i) every point of a HEALPix cell at order k lies within maxpixrad(k) of that cell's centre, and (ii) every angle computed by the classification carries an absolute error below ϵ, then every order-29 cell that intersects R lies in one of the ranges of C.

Proof sketch. The covering starts from cells that tile the whole sphere and, at each step, keeps a cell, drops it, or replaces it by its four children (which tile the parent). The covered area therefore shrinks only when a cell is dropped, which happens only when it is classified outside. By (i), every point of cell (k, p) lies within R_{cell} of its centre, so with (ii) and the triangle inequality, $\angle(\mathbf{q}, \mathbf{c}_{k,p}) - R_{\text{cell}} > r + \epsilon$ guarantees that the whole cell misses the cone; for a polygon the same argument applies to each bounding half-space. Stopping early (range cap, area guard, step limit) keeps cells whole and so cannot drop one either. □

Condition (i) is a property of HEALPix (verified over 1.6×10^6 sampled boundary points, largest observed ratio 0.9999), and (ii) is an assumption about floating-point arithmetic, with ϵ five orders of magnitude above the plausible error.

Stored regions (GiST).. The GiST path relies on three properties of its keys. First, each region's key contains the region. The box key is the exact axis-aligned bound of a spherical cap around the region: the circle itself, or for a polygon the cap centred on the mean of its vertices that reaches its farthest vertex. That cap contains the whole polygon when its radius is at most 90°, as it is for any practical footprint, because such a cap contains the shortest arcs between its points. The box's bounds are rounded outward when stored in single precision, so rounding can only enlarge it. Second, each internal entry's key is the union of its children's keys, so it contains every region below it. Third, a subtree is skipped only when its key rules out a match. For overlap and for containment between regions, the test is whether the boxes overlap: two regions that overlap, or where one contains the other, share a point, so their boxes must overlap too. For containment of a position, the test is whether the box contains the point. Every candidate is then rechecked with the exact predicate, so a loose key costs extra candidates, never missed rows. The cap key works the same way: its caps cover the region, containment is pruned with the overlap test, and every candidate is rechecked.

Tests.. Because these arguments rest on implementation details, the code is also checked exhaustively: the HEALPix layer (known centres, round trips at all 30 orders, nesting, equal area); the covering against a 2-million-point clustered catalogue for cones up to 120° and convex polygons, 30 000 adversarial cones (poles, zone boundaries, face edges, radii 0.02 mas–10°) and 7266 adversarial polygons with 869 117 points inside, all with zero false negatives. SQL regression suites, run on PostgreSQL 16 and 18, require every indexed answer to equal a

Table 1: Which `skycell` index serves which archive workload.

workload	index and query form
cone searches, polygon and box selections on a catalogue	B-tree on `skycell_ang2cell(ra, dec)`; `point <@ circle`, `<@ polygon`, `<@ box`
cross-matching an uploaded target list	the same B-tree; `point <@ circle` in the join, or `skycell_join` (Q3C's argument list)
footprints in a `skyregion` column	GiST, box key by default; cap key (`skyregion_gist_ops`) when region sizes vary widely
which footprints contain a position	the GiST above, or MOC cells in a B-tree or GIN index

Table 2: What a translator emits for the ADQL constructs where the pgSphere-plus-Q3C pair and `skycell` differ most.

ADQL	pgSphere + Q3C	skycell
`POINT`/`CIRCLE` /`POLYGON`	`spoint`/`scircle`/`spoly`	`point`/`circle`/ `polygon`
`CONTAINS(` `POINT,` `CIRCLE)`	`q3c_radial_query` *or* `pos <@ scircle`	`point <@ circle`
`INTERSECTS(` r_1,r_2)	r_1 && r_2 (no Q3C region type)	r_1 && r_2
cross-match	`q3c_join`, a separate function	`point <@ circle` in the join

sequential scan with the exact predicate, for cones, polygons, cross-matches, stored regions and the GiST operator classes, and require the custom scan to return the same rows as the range rewrite.

4. Using skycell in an archive

`skycell` is an ordinary PostgreSQL extension: it adds types, functions, operators and index support, and works on existing tables. A catalogue with `ra` and `dec` columns needs no schema change, only an expression index and fresh statistics:

```
CREATE EXTENSION skycell;
CREATE INDEX cat_cell
  ON cat (skycell_ang2cell(ra, dec));
ANALYZE cat;

-- cone search
SELECT * FROM cat
WHERE point('ICRS', ra, dec)
   <@ circle('ICRS', 266.4, -28.9, 0.05);

-- cross-match against a target list
SELECT * FROM targets t JOIN cat c
  ON point('ICRS', c.ra, c.dec)
  <@ circle('ICRS', t.ra, t.dec, 1/3600.);
```

Storing the cell key as a column and ordering the table by it is optional; it lets nearby sky share heap pages, which the benchmark tables use. Because the covering reads the histogram, `ANALYZE` should be re-run after bulk loads, and a statistics target of 1000 suits catalogues of 10^7 rows or more. A plain B-tree is cheap to maintain: one 8-byte key per row, and `ANALYZE` on a 10-million-row table takes about a second. Table 1 summarises which index serves which workload.

The functions carry ADQL's own names (`point`, `circle`, `polygon`, `contains`, `distance` and the rest), so a TAP service can translate ADQL almost word for word (Table 2), where pgSphere needs its own types and Q3C a separate function per shape. The operator `point <@ circle` is the indexable form of `CONTAINS`, and it uses the custom scan whether the circle is a constant or comes from another table, so an ADQL cross-match is as fast as the native one. ADQL's other spelling of the same search, a `DISTANCE(p1, p2) < r` comparison, is planned as the same cone: the planner adds the cone next to the comparison, which still decides. It also does not matter which way round the query is written. A constant point inside a circle around each row, as the Gaia archive's own examples write cone searches, is planned as the cone around that point, and a cross-match always probes the indexed table, the larger one if both are indexed, whichever table gives the point and whichever the circle. With pgSphere and Q3C the user has to get this right: written the other way round, their cone becomes a full table scan and their cross-match probes the large table's rows one by one against the small one. Regions are written and read as IVOA STC-S text (Rots, 2007), the format `s_region` columns already use. Code written for Q3C can move by renaming: `skycell_radial_query` and `skycell_join` take Q3C's arguments unchanged. The IVOA UDF registry's epoch-propagation functions are also included (ESA, 1997).

5. Performance

5.1. What was measured

The tests use three catalogues of 10 million rows. The *designed* catalogue is synthetic, with a density contrast like that of an optical survey (35% uniform, 45% Galactic disk, 12% bulge, 8% in 200 clusters). The *Gaia* catalogue places positions in proportion to the Gaia DR3 source counts in every order-9 HEALPix cell (3 145 728 cells of 0.11°), reproducing the real sky's crowding down to that scale; a 50-million-row version tests scale. The *real* catalogue is ten million actual Gaia DR3 positions, with clustering at every scale. An ObsCore-shaped table of observations grouped by field, at three sizes, stands for an archive's observation table.

Each method indexes the same rows, stored in the order of its own key. Each query is answered by all methods back to back in random order, five times, and the ratio between methods is taken per query, so the query's own difficulty cancels out. Intervals are 95% bootstrap intervals (4000 resamples), and a difference is reported only when the interval excludes 1. *Warm* results follow an untimed pass; *cold* results are the first touch after a server restart and a page-cache drop. Cone radii run from 1″ to 3°, with 16–400 queries each, half centred on catalogue sources and half placed uniformly on the sky. All runs use PostgreSQL 18.6, Q3C 2.0.5 and pgSphere 1.5.2 with one configuration (2 GB `shared_buffers`, `random_page_cost` 1.1, JIT and parallelism off). Cones, scale, cross-matches, the observation table and the density estimate were measured on a 4-vCPU x86-64 cloud virtual machine with local flash storage (a cold page read costs 0.06–0.1 ms); polygons, stored regions, the

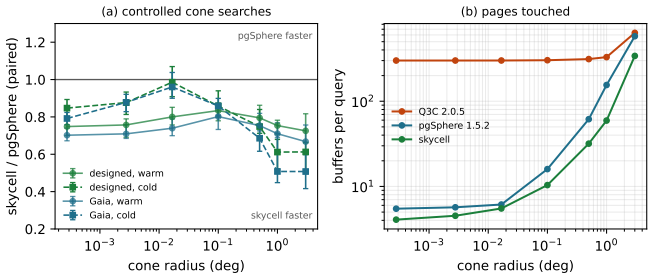


Figure 1: *(a)* Paired ratio of `skycell` to pgSphere against cone radius, 95% bootstrap intervals, synthetic catalogues, warm and cold; below the line `skycell` is faster. *(b)* Buffer pages touched per query (mean, designed catalogue).

memory test and the TAP service on an Apple M3 Pro laptop under Docker.

5.2. Cone searches

Cone searches are the most common archive query and pgSphere's strength. Table 3 and Fig. 1 show that, with a warm cache, `skycell` is faster than pgSphere at every radius on all three catalogues: by 17–33% on the two synthetic ones and by 4–31% on real positions. At 1″ it spends slightly longer planning (0.036 ms against 0.029) but executes faster (0.022 ms against 0.040), reading 4.1 pages to pgSphere's 5.5. Cold, it is faster or level everywhere, and the margin grows with radius to a factor 1.6–2.1 at 1° and 3°, because it touches about a third of pgSphere's pages there (59 against 155 on the designed catalogue, 85 against 282 on the Gaia one). On real positions the cold gain is a factor 1.5–2.1 from 6′ up; below that a first query reads only two or three pages and the two methods are about level. These cold numbers come from fast flash storage, where a saved page is worth about 0.1 ms; on a slower disk each saved page counts for more.

Against Q3C, `skycell` takes 0.02–0.58 of the time. This mostly reflects Q3C 2.0.5's implementation rather than its design: `q3c_radial_query` expands every call into about 100 bitmap index scans regardless of radius, and spends about 2.7 ms planning them. Q3C's cross-match, which does not have this overhead, is the fairer comparison (Section 5.4).

5.3. Catalogue size, index size and build time

Ten million rows is 0.55% of Gaia DR3, so the tests were repeated at 50 million rows. At that size pgSphere's index (3439 MB) no longer fits in the 2 GB buffer pool, while `skycell`'s (1071 MB) still does (Table 4). With a warm cache `skycell` stays 14–31% faster at every radius. Cold, its lead grows with catalogue size from 30′ up, to a factor 1.9 at 30′, 2.7 at 1° and 2.4 at 3°, as the gap in pages read widens (188 against pgSphere's 642 at 1°). Its row estimates, which the planner uses to choose join strategies, stay within a factor 1.3 of the truth, while pgSphere's are off by up to a factor 8.7 and Q3C's by up to 2.4.

Index size and build time decide how much memory a server needs and how long a new catalogue release takes to index. The cost per row is flat between 10^7 and 5×10^7 rows, so it can be extrapolated to Gaia DR3's 1.81×10^9 sources (Table 5). The relevant comparison is between deployments: an archive that serves both cone searches and cross-matches today typically runs pgSphere plus Q3C, whose `q3c_join` is 2.4–2.6 times faster than pgSphere at cross-matching (Section 5.4). `skycell` serves both from one index. At DR3 scale that is 38 GiB against 160 GiB, the difference between a 64 GiB and a 256 GiB server, and an index build of 10 minutes against 4.1 hours. The smaller index also helps when memory is short: in a 3 GiB container with 768 MB of shared buffers and a 22 GB, 20.5-million-row ObsCore-shaped table, where pgSphere's index is 1.5 times the buffer pool and `skycell`'s 0.57 times, `skycell` answered a stream of 2° cones across the sky 4.2–4.8 times faster.

5.4. Cross-matching and polygons

Cross-matching an uploaded target list is Q3C's speciality, so it is the main test of the B-tree design against Q3C's. On real Gaia DR3 positions, at the radii cross-matching uses (1–1.5″ for optical, up to 5″ for radio) and with uploads of 10^3–10^5 targets, clustered or uniform (Table 6), `skycell` takes 0.79 of `q3c_join`'s time and 0.32 of pgSphere's overall, and is faster than both in all 24 test cells. On the resampled catalogue, with 200 000 targets at 1″ and 10″, it takes 0.67–0.75 of `q3c_join`'s time and 0.26–0.31 of pgSphere's, with identical results. The gain comes from covering each target's cone with only the ranges it needs, 1.2 on average at 1″ (4.3 pages per target). The join can be written with `skycell_cone`, `skycell_join` or ADQL's `point <@ circle`; all three produce the same plan.

Convex polygons of 0.05°–2° take a median 0.28 ms against pgSphere's 0.77 and Q3C's 1.86, a factor 2.7–4.4 on both synthetic catalogues; pgSphere's slowest queries are far above its median when a polygon's bounding volume fits it poorly.

5.5. An observation table

Observation tables cluster around field centres and are often ordered by collection, not position. An ObsCore-shaped table, with observations in groups of 128 sharing a field centre, was built at three sizes and asked the same 30 queries at each (Table 7). `skycell`'s lead grows with table size: at 10^7 rows it is level on the 0.15° cone and faster in every other class, by 25% on the 2° cone, where pgSphere reads 939 pages to its 591. pgSphere wins only the small cone on the smallest table, where its bitmap scan already reads exactly the pages needed. On tables that small the index choice matters little anyway: inside a TAP 1.1 service (Delli Veneri & Salgado, 2026) on a 500 000-row ObsCore table, the two indexes gave the same throughput at 1–32 concurrent clients, because the service's own code costs about 10 ms per request against a fraction of a millisecond in the database.

5.6. How reliable is the density estimate?

Table 8 compares the density estimate of Eq. (6) with the density counted from the real-position catalogue. It is within 5% in Baade's Window and a factor 1.3 at the Galactic pole, but low by a factor 10–50 in globular cluster cores and high by

Table 3: Cone searches: paired ratio of `skycell` to pgSphere with its 95% bootstrap interval. Below 1, `skycell` is faster. **Bold**: interval excludes 1. *n*: distinct queries at that radius. Real catalogue: its own centres at the same radii and counts; cold there uses one table per method and fresh centres, so neither method warms the other's pages.

		designed catalogue		Gaia catalogue (resampled)		real Gaia DR3 positions	
radius	*n*	warm	cold	warm	cold	warm	cold
1″	400	**0.75** [0.70, 0.79]	**0.85** [0.80, 0.89]	**0.70** [0.67, 0.74]	**0.79** [0.76, 0.83]	**0.83** [0.81, 0.84]	**0.87** [0.82, 0.94]
10″	400	**0.76** [0.69, 0.79]	**0.88** [0.81, 0.94]	**0.71** [0.68, 0.76]	**0.88** [0.83, 0.92]	**0.85** [0.82, 0.86]	1.01 [0.94, 1.09]
1′	300	**0.80** [0.76, 0.85]	0.99 [0.91, 1.07]	**0.74** [0.70, 0.79]	0.96 [0.90, 1.04]	**0.90** [0.87, 0.92]	0.98 [0.90, 1.04]
6′	200	**0.83** [0.78, 0.88]	**0.86** [0.80, 0.94]	**0.80** [0.74, 0.86]	**0.86** [0.81, 0.90]	**0.96** [0.93, 0.99]	**0.61** [0.55, 0.65]
30′	100	**0.79** [0.75, 0.86]	**0.75** [0.71, 0.84]	**0.75** [0.73, 0.82]	**0.69** [0.62, 0.76]	**0.81** [0.78, 0.86]	**0.67** [0.60, 0.72]
1°	40	**0.75** [0.71, 0.78]	**0.61** [0.52, 0.68]	**0.71** [0.69, 0.77]	**0.51** [0.45, 0.72]	**0.80** [0.75, 0.88]	**0.53** [0.44, 0.66]
3°	16	**0.73** [0.66, 0.82]	**0.61** [0.50, 0.66]	**0.67** [0.63, 0.76]	**0.51** [0.41, 0.60]	**0.69** [0.64, 0.75]	**0.48** [0.43, 0.54]

Table 4: Paired ratio of `skycell` to pgSphere on the Gaia density field at two catalogue sizes. **Bold**: interval excludes 1.

	warm		cold	
radius	10^7	$5{\times}10^7$	10^7	$5{\times}10^7$
1″	**0.70**	**0.78**	**0.79**	**0.82**
10″	**0.71**	**0.78**	**0.88**	**0.86**
1′	**0.74**	**0.86**	0.96	0.98
6′	**0.80**	**0.86**	**0.86**	**0.81**
30′	**0.75**	**0.75**	**0.69**	**0.53**
1°	**0.71**	**0.69**	**0.51**	**0.37**
3°	**0.67**	**0.73**	**0.51**	**0.42**

Table 5: Index size and build time at 10^7 rows, and extrapolated linearly from the 5×10^7 measurement to Gaia DR3 (1.81×10^9 sources). Last column: the smallest conventional server memory holding the index with room for the table.

	10^7 rows		at Gaia DR3 scale		
index	MB	build (s)	GiB	build	mem GiB
Q3C	214	5.6	38	15 min	64
pgSphere	685	68.6	122	3.9 h	192
skycell	214	6.0	**38**	10 min	64
pgSphere + Q3C	—	—	160	4.1 h	256

Table 6: Cross-matching on real Gaia DR3 positions (10^7 rows): geometric mean over three upload sizes (10^3–10^5) and two target distributions of the per-cell ratio of median times, method order random per cell. *Cone*: the join written with `skycell_cone`; *join*: with `skycell_join`, Q3C's spelling. Below 1, `skycell` is faster; **bold**: faster in every cell.

	vs `q3c_join`		vs pgSphere
radius	cone	join	join
0.5″	**0.81**	**0.77**	**0.30**
1″	**0.77**	**0.76**	**0.30**
1.5″	**0.77**	**0.78**	**0.34**
5″	**0.84**	**0.83**	**0.35**
all	**0.80**	**0.79**	**0.32**

Table 7: ObsCore-shaped table at three sizes: total query time of `skycell` as the ratio to pgSphere (pgS.), 30 stored centres, two complete randomized runs pooled; median buffer pages at 10^7 rows. Q05: 0.15° cone; Q06: 2°; Q07: 0.5° with calibration and time cuts; Q12: an empty 0.7″ cone.

	`skycell` / pgS.			buffers, 10^7	
rows (10^6)	0.5	2	10	pgS.	skycell
Q05	1.14	1.07	0.97	32	32
Q06	0.92	0.93	0.75	939	591
Q07	0.94	0.99	0.80	113	89
Q12	0.71	0.68	0.47	7	3

Table 8: The density estimate $\hat{\rho}$ against the true density ρ on real Gaia DR3 positions, radii 0.01°–1°: median ratio over radii, then over ten `ANALYZE` samples; *samples*: spread of that median across them; *range*: over all radii and samples.

field	$\hat{\rho}/\rho$	samples	range
ω Cen	**0.08**	0.06–0.09	0.05–0.54
47 Tuc	**0.02**	0.01–0.11	0.00–0.94
LMC	0.80	0.46–0.92	0.22–1.26
Baade's Window	0.96	0.92–0.98	0.45–1.19
Galactic pole	1.27	1.18–1.49	0.48–1.69
Galactic centre	**3.71**	3.41–3.80	1.95–5.20

Table 9: Region operators on the 50 000-row `fpr` catalogue, 500 probes per operator: time of the box-key GiST index as the ratio to pgSphere's native GiST. Below 1, `skycell` is faster.

operator	box key / pgSphere
`&&` (overlap)	**0.12–0.17**
`@>`(region,point)	**0.75–0.93**
`@>`(region,region)	**0.55–0.70**
`<@`(region,region)	**0.26–0.27**

3.7 at the Galactic centre: the histogram resolves a cluster while the query sits inside it, but averages it away once the query is larger. The effect on queries is much smaller than the error. Since $s_\star \propto \rho^{-1/2}$, an error of ten in the density changes the cell size by about three, and the cost curve is flat over much of that range. In cluster cores the covering chosen is within 1.05–1.53 of the best fixed cell size, and within 0.98–1.31 elsewhere, whereas a badly chosen fixed size is 243–830 times worse.

5.7. Stored regions

The GiST index was tested on the real 50 000-row `fpr` footprint catalogue against pgSphere's native GiST index. With the default box key, `skycell` is faster for all four region operators (Table 9), from 6–8 times faster for overlap to 7–25% faster for finding the footprints that contain a position. On footprints of a single size it reads 10–25% fewer buffer pages in 10–20% less time, and on a column mixing circles and polygons the margin widens, because pgSphere has to search one index per region type. For the point-in-footprint query, storing footprints as MOC cells in a B-tree answered 200 474 point-in-region pairs in 0.76 s against pgSphere's 0.97 s.

6. Conclusions

`skycell` keeps Q3C's compact integer key and B-tree, chooses each covering's resolution from local source density

and the cost of an index range, and adds a GiST index for stored regions. Compared directly with the two extensions archives use today, it is faster than each at its own speciality: cone searches are faster than with pgSphere at every radius from 1″ to 3° with a warm cache (by 17–33% on synthetic catalogues and 4–31% on real Gaia DR3 positions), faster or level when cold, and further ahead as the catalogue grows; cross-matches are 20–33% faster than `q3c_join` and about three times faster than pgSphere. Convex polygons run 2.7–4.4 times faster than with pgSphere, and the region index is faster than pgSphere's on every operator tested.

For an archive the operational gains may matter as much: one index, a third the size of pgSphere's and built 11–23 times faster, replaces the pgSphere-plus-Q3C pair, with nothing to tune. Adoption needs only an expression index on existing `ra` and `dec` columns, and queries keep ADQL's own names. This makes `skycell` a strong candidate for the positional layer of a PostgreSQL archive or TAP service.

The gains are smallest on small tables, where planning a covering is a larger share of each query and a small cone on 5×10^5 rows can still favour pgSphere; there, as inside a TAP service dominated by its own code, an existing index is good enough.

CRediT authorship contribution statement

Jesús Salgado: Conceptualization, Methodology, Software, Formal analysis, Writing – original draft. **Michele Delli Veneri:** Validation, Investigation, Writing – review & editing.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Declaration of generative AI and AI-assisted technologies in the writing process

During the preparation of this work the authors used Claude Code (Anthropic) in order to develop the software, the benchmark harness and the analysis, and to draft the manuscript. After using this tool, the authors reviewed and edited the content as needed and take full responsibility for the content of the published article. The geometric predicates and coverings are checked by the brute-force suites of Section 3 against independent computations over millions of adversarially placed points rather than expected outputs, and every indexed answer in the regression suites must equal a sequential scan with the exact predicate.

Data availability

The extension, its test suites, the benchmark harness and the raw results of every measurement quoted here are available at https://github.com/jesusjuansalgado/skycell under the MIT licence, together with the history of the measurements that led to the current defaults. All corpora but the real one are generated from seeds; the real one is drawn from the Gaia archive by a recorded query.

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